

TrES-3b

R-2146

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_130225 · created 2026-08-12T00:41:30+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456348.9561 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.148 +/- 0.035
Transit depth (Rp/Rs)^2	2.2 +/- 1.02 [%]
Orbital Inclination (inc)	82.37 +/- 1.38
Airmass coefficient 1 (a1)	0.967 +/- 0.015
Airmass coefficient 2 (a2)	0.0165 +/- 0.0082
Scatter in the residuals of the lightcurve fit is	1.5 %
Best Comparison Star	#3 - [312, 361]
Optimal Method	PSF photometry
Transit Duration (day)	0.055 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.148 ± 0.035 vs 0.16309 (deviation 0.43σ)
Duration fit vs published	0.055 d vs 0.0567 d (deviation 0.11σ)
Mid-time O-C	-0.0 min
Depth SNR	4.2
Residual scatter	1.5 %

Light curve

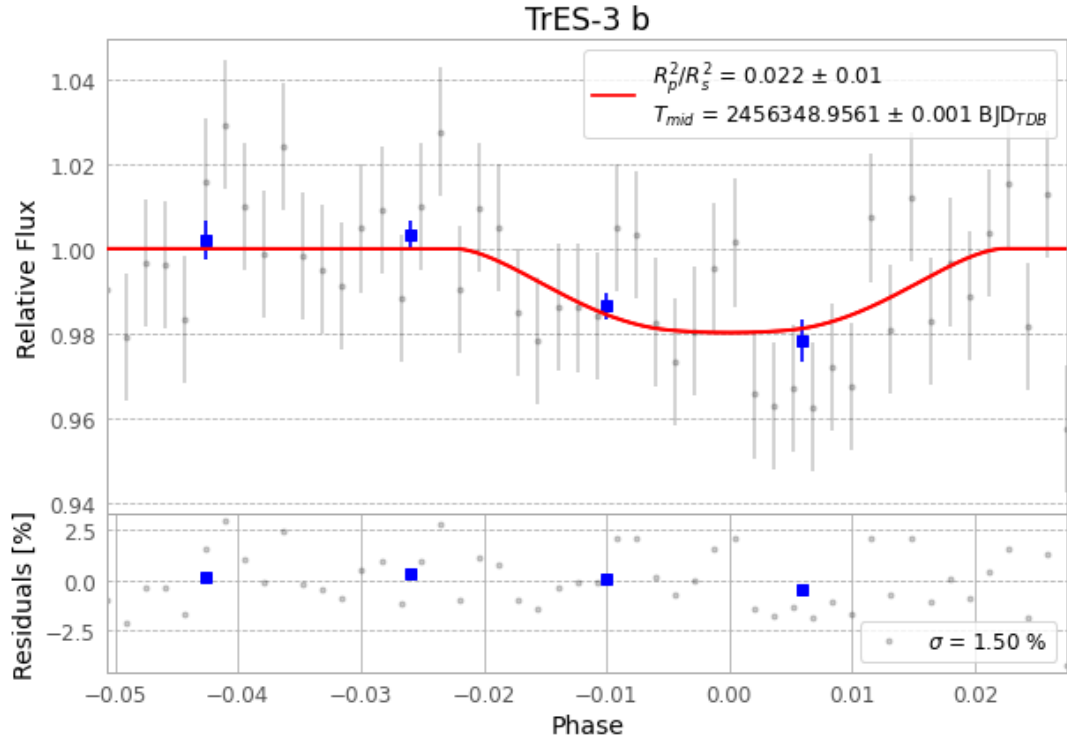


Figure 1 — FinalLightCurve_TrES-3 b_2013-02-25.png (SHA-256 4445678891e47874...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [265, 150], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 76f222204b47485c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

50 FITS frames (33 MB), TRES-3130225092113.FITS → TRES-3130225114812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-130225.log (1371d2bcb838...), FinalLightCurve_TrES-3 b_2013-02-25.png (4445678891e4...), FinalLightCurve_TrES-3 b_2013-02-25.csv (713f5546a452...)

Compute resources

Wall time 131.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-12 00:41Z.